

AP EAMCET-2019-Engineering-20 Apr-Shift-2
Questions Mathematics

1. If $f : \mathbb{R} - \left\{ \frac{3}{7} \right\} \rightarrow \mathbb{R} - \left\{ \frac{3}{7} \right\}$ is given by $f(x) = \frac{3x+5}{7x-3}$, then the

statement which is not true is

(1) $f^{-1}(x) = f(x)$

(2) $(f \circ f)(x) = x$

(3) $(f \circ f \circ f)(x) = x$

(4) $(f \circ f \circ f \circ f)(x) = x$

2. If $f : A \rightarrow B$, $g : B \rightarrow C$ are functions such that $g \circ f : A \rightarrow C$ is onto, then a necessary condition is

(1) f is onto

(2) g is onto

(3) both f, g are onto

(4) neither f nor g is onto.

3. The sum of the first n terms of the series $\frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots$ is

$$(1) \ n + \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3}$$

$$(2) \ n - \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3}$$

$$(3) \ n + \frac{2^{n+1}}{3 \times 5^n} + \frac{2}{3}$$

$$(4) \ n - \frac{2^{n+1}}{3 \times 5^n} + \frac{2}{3}$$

4. The sum of the values of x so that the matrix

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is singular is,}$$

$$(1) \ 3$$

$$(2) \ 5$$

$$(3) \ 7$$

$$(4) \ 9$$

$$5. \text{ If } A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ then } (\text{Adj } A)^{-1} =$$

- (1) $A + I$
- (2) $A - I$
- (3) A
- (4) $\text{Adj}(A^{-1})$

6. If $abc \neq 0$ and the system of equation $x + 7ay + 2az = 0$, $a + 6by + 2bz = 0$, $x + 5cy + 2cz = 0$ has a non-trivial solution, then a, b, c are in

- (1) Harmonic Progression
- (2) Geometric Progression
- (3) Arithmetic Progression
- (4) Arithmetic-geometric Progression

7. z_1, z_2 are complex numbers with $|z_1 - z_2| < k$. If the complex number z satisfies the condition $|z - z_1| + |z - z_2| = k$, then z lies on

- (1) a parabola
- (2) an ellipse
- (3) a circle
- (4) a hyperbola

8. In a complex plane $\mathbb{C}$, the set $\left\{ z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4} \right\}$

represents

- (1) a straight line
- (2) a circle
- (3) a parabola
- (4) an ellipse

9. For $a, b, c, d \in \mathbb{R}$, if $z_1 = a + ib$, $z_2 = c + id$ are such that $|z_1| = |z_2| = 1$ and $\operatorname{Re}(z_1 \bar{z}_2) = 0$, then the pair of complex numbers $w_1 = a + ic$ and $w_2 = b + id$ satisfy

- (1) $\operatorname{Re}(w_1 \bar{w}_2) = 0$
- (2) $\operatorname{Re}(w_1 \bar{w}_2) = 1$
- (3) $|w_1| \neq |w_2|$
- (4) $|w_1| = |w_2| = 0$

10. If $(1+x)^n = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and

$$a_0 - a_2 + a_4 - a_6 + \cdots = k \cos \frac{n\pi}{4}, \text{ then } k =$$

- (1) 2^n
- (2) 2^{2n}
- (3) $\frac{2^n}{2}$
- (4) $2^{n/2}$

11. For $x > 2$, the equation $\sqrt{x+2} - \sqrt{x-2} = \sqrt{4x-2}$ has

- (1) One Solution
- (2) Two solutions
- (3) More than two solutions
- (4) No solution

12. If $x \in \mathbb{R}$, then the range of $\frac{x}{x^2 - 5x + 9}$ is

- (1) $\left(-\frac{1}{11}, 1\right)$
- (2) $\left(-\infty, \frac{-1}{11}\right) \cup (1, \infty)$

$$(3) \left[-\frac{1}{11}, 1 \right]$$

$$(4) \left[-1, \frac{1}{11} \right]$$

13. If α, β, γ are the roots of the equation $x^3 - ax^2 + bx - c = 0$ then

$$\sum \alpha^2 (\beta + \gamma) =$$

$$(1) ab - 3c$$

$$(2) \frac{ab - 3c}{c}$$

$$(3) \frac{b^2 - 2ac}{c^2}$$

$$(4) \frac{a^2 - 2b}{c^2}$$

14. The sum of all the rational roots of the equation

$$6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0 \text{ is}$$

$$(1) 3$$

$$(2) 3.5$$

$$(3) \frac{25}{6}$$

(4) 2.5

15. The number of five digit numbers that are divisible by 6 which can be formed by choosing digits from $\{0,1,2,3,4,5\}$, when repetition is allowed, is

(1) 648

(2) 540

(3) 1296

(4) 1080

16. A person invites 8 guests to a dinner and places 5 of them at one table and the remaining 3 at another, both the tables being round. The number of ways in which the guest can be arranged is

(1) 40320

(2) 2688

(3) 8064

(4) 13440

17. If 15^k divides $47!$ but 15^{k+1} does not divide it, then $k =$

- (1) 15
- (2) 12
- (3) 10
- (4) 5

18. If $(1 + x + x^2)^n = c_0 + c_1x + c_2x^2 + \dots$, then the value of $c_0c_1 - c_1c_2 + c_2c_3 - \dots$ is

- (1) $(-1)^n$
- (2) 0
- (3) 2^n
- (4) 3^n

19. If x is so small that x^2 and higher power of x can be neglected,

then the approximate value of $\left(1 + \frac{3}{4}x\right)^{\frac{1}{2}} \left(1 - \frac{2x}{3}\right)^{-2}$ is

- (1) $\frac{41 + 24x}{41}$
- (2) $\frac{41 - 24x}{41}$

$$(3) \frac{24 + 41x}{24}$$

$$(4) \frac{24 - 41x}{24}$$

20. If $\frac{x^4}{(x-a)(x-b)(x-c)} = P(x) + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$, then

$$P(0) + A(a-b)(a-c) =$$

(1) $a^4 + b^4 + c^4 + a$

(2) $a + b + c$

(3) $a^4 - a - b - c$

(4) $a + b + c + a^4$

21. If $\sin \alpha = p$, then the quadratic equation whose roots are $\tan \frac{\alpha}{2}$,

$\cot \frac{\alpha}{2}$ is

(1) $px^2 - 2x + p = 0$

(2) $px^2 + 2x + p = 0$

(3) $px^2 + x + p = 0$

(4) $px^2 - x + p = 0$

22. $\cos 66^\circ + \sin 84^\circ =$

(1) $\frac{1}{4}(\sqrt{3} + \sqrt{5})$

(2) $\frac{1}{4}\sqrt{5}(\sqrt{3} + 1)$

(3) $\frac{1}{4}(\sqrt{3} + 1)(\sqrt{5} + 1)$

(4) $\frac{1}{4}\sqrt{3}(\sqrt{5} + 1)$

23. If $\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$, then the general value of θ are

(1) $(3n+1)\frac{\pi}{3}, n \in \mathbb{Z}$

(2) $(3n+1)\frac{\pi}{9}, n \in \mathbb{Z}$

(3) $(3n+1)\frac{\pi}{6}, n \in \mathbb{Z}$

(4) $(2n+1)\frac{\pi}{9}, n \in \mathbb{Z}$

24. The sum of the values of x satisfying the equation

$$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}(x), \text{ is}$$

(1) 0

(2) $\frac{1}{2}$

(3) $\frac{1}{3}$

(4) $\frac{1}{4}$

25. If $\theta = \frac{\pi}{6}$ and $x = \log\left[\cot\left(\frac{\pi}{4} + \theta\right)\right]$ then $\sinh(x) =$

(1) $\sqrt{3}$

(2) $\frac{1}{\sqrt{3}}$

(3) $-\sqrt{3}$

(4) $-\frac{1}{\sqrt{3}}$

26. In ΔABC , if the median AD drawn through A is perpendicular to the side AC , then $3ca \cos A \cos C + 2a^2 =$

- (1) c^2
- (2) $2c^2$
- (3) $3c^2$
- (4) $4c^2$

27. Let Δ denote the area of a ΔABC . If α, β, γ are the lengths of the altitudes of the ΔABC , then $\alpha^{-2} + \beta^{-2} + \gamma^{-2} =$

- (1) $\frac{4}{\Delta}(\tan A + \tan B + \tan C)$
- (2) $\frac{1}{\Delta}(\cot A + \cot B + \cot C)$
- (3) $\frac{\Delta^2}{2}(\tan A + \tan B + \tan C)$
- (4) $\frac{\Delta^2}{4}(\cot A + \cot B + \cot C)$

28. In ΔABC , $(r_2 + r_3) \cot\left(\frac{B+C}{2}\right) =$

- (1) $a + b + c$

(2) a

(3) b

(4) c

29. If the vectors $\overrightarrow{AB} = \vec{i} + 3\vec{j} + 4\vec{k}$, $\overrightarrow{AC} = 5\vec{i} + \vec{j} + 2\vec{k}$ are two sides of triangle ABC, whose centroid is G, then $|\overrightarrow{AG}| =$

(1) $\frac{2}{3}\sqrt{22}$

(2) $\sqrt{22}$

(3) $\frac{1}{3}\sqrt{22}$

(4) $\sqrt{18}$

30. The point of intersection of the lines represented by

$$\vec{r} = (\vec{i} + 2\vec{j} - \vec{k}) + \lambda(2\vec{i} + 3\vec{j} + 4\vec{k}) \quad \text{and}$$

$$\vec{r} = (-\vec{i} - 3\vec{j} + 7\vec{k}) + \mu(\vec{i} + 2\vec{j} - \vec{k}) \text{ is}$$

(1) $3\vec{i} + 5\vec{j} + 3\vec{k}$

(2) $5\vec{i} + 8\vec{j} + 7\vec{k}$

(3) $-\vec{i} - \vec{j} - 5\vec{k}$

(4) $-3\bar{i} - 4\bar{j} - 9\bar{k}$

31. If A, B, C and D are points whose position vectors are $\bar{i} + \bar{j} + \bar{k}$, $4\bar{i} - \bar{j} + 2\bar{k}$, $5\bar{i} + \bar{j}$, $7\bar{i} + 2\bar{j} + 3\bar{k}$ respectively, then the projection of $\overline{AB}$ on $\overline{CD}$ is

(1) $\frac{4}{3}$

(2) $\sqrt{\frac{2}{7}}$

(3) $\frac{3}{4}$

(4) $\sqrt{\frac{7}{2}}$

32. Let $\bar{a}$ and $\bar{b}$ be unit vectors with θ as the acute angle between

them. If $\frac{1}{2}|\bar{a} - \bar{b}| = \sin \lambda \theta$, then $4\lambda^2 =$

(1) 4

(2) 1

(3) 3

(4) 2

33. Let $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$ and $\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$. The volume(in cubic units) of the parallelopiped having $\vec{a} + \vec{b} + \vec{c}$, $\vec{a} - \vec{b} + \vec{c}$ and $\vec{a} + \vec{b} - \vec{c}$ as coterminal edges is

- (1) 6
- (2) 7
- (3) 28
- (4) 36

34. If $|\vec{a}| = 1$, $|\vec{b}| = 1$, $|\vec{c}| = 2$ and $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$, then $(\vec{a} \cdot \vec{c})^2 =$

- (1) 1
- (2) 2
- (3) 4
- (4) 3

35. The variance of the observations 2,3,5,7,11,13,17,22 is

- (1) 43.75
- (2) 48.25
- (3) 80
- (4) 350

36 The coefficient of variation and standard deviation of an ungrouped data are 60 and 21 respectively. If 15 is added to every observation of the data, then the coefficient of variation of the new data is

- (1) 30
- (2) 42
- (3) 40
- (4) 20

37. Each of the two boxes A and B contains 10 chits numbered 1 to 10. If one chit is drawn at random from each of A and B, then the probability that the number on the chit drawn from A is smaller than the number on the chit drawn from B, is

- (1) $\frac{9}{10}$
- (2) $\frac{9}{20}$
- (3) $\frac{19}{20}$

(4) $\frac{17}{20}$

38. An envelope is known to have come from either 'LONDON' or 'CLIFTON'. On the postal mark only two successive letters ON are legible. The probability that the envelope comes from LONDON is

(1) $\frac{12}{17}$

(2) $\frac{5}{17}$

(3) $\frac{3}{17}$

(4) $\frac{2}{5}$

39. The probability function of a random variable X is given by $P(X = k) = ck^2$, where c is a constant and $k \in \{0, 1, 2, 3, 4\}$. If σ^2 is the variance of X and μ is the mean of X , then $\sigma^2 + \mu^2 =$

(1) 3.33

(2) 11.8

(3) $\frac{1}{30}$

(4) 354

40. In a book of 250 pages, there are 200 typographical errors. Assuming that the number of errors per page follow the Poisson law, then the probability that a random sample of 5 pages will contain no typographical error is

(1) e^{-4}

(2) $e^{-\frac{4}{5}}$

(3) $e^{-\frac{16}{25}}$

(4) e^{-1}

41. The locus of the point $P(x, y)$ satisfying the equation

$$\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4 \text{ is}$$

(1) an ellipse

(2) a parabola

(3) a line segment

(4) a circle

42. If the origin is shifted to $(2,3)$ and the axes are rotated through an angle 45° about that point, then the transformed equation of $2x^2 + 2y^2 - 8x - 12y + 18 = 0$ is

(1) $x^2 - 7y^2 - 14xy - 2 = 0$

(2) $x^2 + y^2 = 4$

(3) $x^2 - y^2 = 4$

(4) $8x^2 - 2y^2 = 9$

43. A line passing through $P(4,2)$ cuts the coordinate axes at A and B respectively. If O is the origin, then the locus of the centre of the circum-circle of $\triangle OAB$ is

(1) $x^{-1} + y^{-1} = 2$

(2) $2x^{-1} + y^{-1} = 1$

(3) $x^{-1} + 2y^{-1} = 1$

(4) $2x^{-1} + 3y^{-1} = 1$

44. The centroid of the triangle formed by the lines $x + y - 1 = 0$, $x - y - 1 = 0$, $x - 3y + 3 = 0$ is

(1) $\left(\frac{4}{3}, 1\right)$

(2) $\left(-\frac{4}{3}, 1\right)$

(3) $\left(\frac{8}{3}, 3\right)$

(4) $\left(\frac{-8}{3}, 3\right)$

45. The quadratic equation whose roots are the coordinates of the circumcentre of the triangle formed by the points $(-2, -1)$, $(6, -1)$, $(2, 5)$ is

(1) $x^2 - 5x + 6 = 0$

(2) $2x^2 - 9x + 9 = 0$

(3) $3x^2 - 8x + 4 = 0$

(4) $6x^2 - 13x + 6 = 0$

46. The straight line $x + y + 1 = 0$ bisects an angle between a pair of lines, of which one is $2x - 3y + 4 = 0$. Then the equation of the other line in that pair is

(1) $2x + 3y + 4 = 0$

(2) $x - y + 1 = 0$

(3) $5x - 5y + 9 = 0$

(4) $3x - 2y + 5 = 0$

47. If the pairs of straight lines represented by $3x^2 + 2hxy - 3y^2 = 0$ and $3x^2 + 2hxy - 3y^2 + 2x - 4y + c = 0$ form a square then

$(h, c) =$

(1) $(4, -1)$

(2) $(-1, 4)$

(3) $(-4, 1)$

(4) $(1, -4)$

48. The equation of the bisectors of the angles between the lines joining the origin to the points of intersection of the curve $x^2 + xy + y^2 + x + 3y + 1 = 0$ and the line $x + y + 2 = 0$ is

(1) $x^2 + 4xy - y^2 = 0$

(2) $2x^2 + 5xy - y^2 = 0$

(3) $x^2 + 6xy - 2y^2 = 0$

(4) $2x^2 - 4xy + 2y^2 = 0$

49. A is the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. If the tangents drawn at the points B(1,7) and D(4,-2) on the circle meet at the point C, the area of the quadrilateral ABCD (in square units) is

(1) 75

(2) 64

(3) 56

(4) 45

50. The point of concurrence of all conjugate lines of the line $5x + 7y - 78 = 0$ with respect to the circle $x^2 + y^2 + 6x + 8y - 96 = 0$ is

(1) (-2,3)

(2) (3,-2)

(3) (3,2)

(4) (2,3)

51. The number of common tangents to the circles

$$x^2 + y^2 + 4x - 6y - 12 = 0 \text{ and } x^2 + y^2 - 8x + 10y + 5 = 0 \text{ is}$$

- (1) 4
- (2) 3
- (3) 2
- (4) 1

52. If two circles $(x-1)^2 + (y-3)^2 = r^2$ and

$$x^2 + y^2 - 8x + 2y + 8 = 0 \text{ intersect at two distinct points, then}$$

- (1) $2 < r < 8$
- (2) $1 < r < 9$
- (3) $r = 2$
- (4) $r = 8$

53. If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$ then $k =$

- (1) 21
- (2) -21
- (3) -23
- (4) 23

54. The equation of the circle which cuts the circles

$$x^2 + y^2 + 4y - 7 = 0,$$

$$2x^2 + 2y^2 + 3x + 5y - 9 = 0,$$

$x^2 + y^2 + y = 0$ orthogonally is

(1) $x^2 + y^2 - 4x - 2y - 1 = 0$

(2) $x^2 + y^2 - 4x - 6y - 3 = 0$

(3) $x^2 + y^2 - 4x - 2y - 3 = 0$

(4) $x^2 + y^2 - 2x - 4y - 1 = 0$

55. The vertex and the focus of the parabola $2y^2 + 5x - 6y + 1 = 0$ are respectively

(1) $\left(\frac{7}{10}, \frac{3}{2}\right), \left(\frac{3}{40}, \frac{3}{2}\right)$

(2) $\left(-\frac{7}{10}, \frac{3}{2}\right), \left(\frac{53}{40}, \frac{3}{2}\right)$

(3) $\left(\frac{7}{10}, -\frac{3}{2}\right), \left(\frac{7}{10}, \frac{7}{8}\right)$

(4) $\left(-\frac{7}{10}, -\frac{3}{2}\right), \left(\frac{7}{10}, \frac{17}{8}\right)$

56. If a normal chord at point $t (\neq 0)$ on the parabola $y^2 = 9x$ subtends a right angle at its vertex then $t =$

- (1) $\sqrt{3}$
- (2) $\sqrt{5}$
- (3) $\pm\sqrt{3}$
- (4) $\pm\sqrt{2}$

57. The major and minor axes of an ellipse are along the X-axis and the Y-axis respectively. If its latus rectum is of length 4 and the distance between the foci is $4\sqrt{2}$, then the equation of ellipse is

- (1) $2x^2 + y^2 = 16$
- (2) $x^2 + 2y^2 = 16$
- (3) $\frac{x^2}{2} + \frac{y^2}{3} = 1$
- (4) $\frac{x^2}{3} + \frac{y^2}{2} = 1$

58. If $c \in \mathbb{R}$ such that the line $4x - y + c = 0$ touches the ellipse $x^2 + 4y^2 = 4$, then an equation having all such values of c among its roots is

$$(1) x^2 - (1 + \sqrt{17})x + \sqrt{17} = 0$$

$$(2) x^2 - (1 - \sqrt{17})x - \sqrt{17} = 0$$

$$(3) x^3 - x^2 - 17x + 17 = 0$$

$$(4) x^3 - x^2 + 17x + 17 = 0$$

59. If e_1 , e_2 are respectively the eccentricities of the curves

$$9x^2 - 16y^2 - 144 = 0 \text{ and } 9x^2 - 16y^2 + 144 = 0 \text{ then } \frac{e_1^2 e_2^2}{e_1^2 + e_2^2} =$$

$$(1) \sqrt{2}$$

$$(2) 1$$

$$(3) \sqrt{3}$$

$$(4) 2$$

60. If $A(2, 4, -1)$, $B(3, 6, -1)$ and $C(4, 5, 1)$ are three consecutive vertices of a parallelogram, then its fourth vertex is

$$(1) (1, 3, 3)$$

$$(2) (1, 3, -3)$$

$$(3) (3, 3, -1)$$

(4) $(3,3,1)$

61. If the line joining the points $A(2,3,1)$ and $B(3,5,-3)$ is perpendicular to the line joining the $C(1,2,3)$ and $D(3,y,7)$ then

$y =$

(1) 1

(2) 3

(3) 5

(4) 7

62. A plane is making intercepts 2,3,4 on X, Y and Z-axes respectively. Another plane is passing through point $(-1,6,2)$ and is perpendicular to the line joining the points $(1,2,3)$ and $(-2,3,4)$. Then, the angle between the two planes is

(1) 90°

(2) $\cos^{-1} \sqrt{\frac{12}{61}}$

(3) $\cos^{-1} \sqrt{\frac{11}{61}}$

(4) $\cos^{-1} \sqrt{\frac{5}{6}}$

63. If $\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$ and $\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ then

(1) $\alpha = \beta$

(2) $2\alpha = \beta$

(3) $\alpha = 2\beta$

(4) $\alpha = 3\beta$

64. If $f(x) = \begin{cases} \frac{x - |x|}{x}, & \text{when } x < 0 \\ 5x^2 + a, & \text{when } 0 \leq x \leq 1 \\ b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right), & \text{when } 1 < x < 3 \\ -14, & \text{when } x \geq 3 \end{cases}$ is a continuous

function on $\mathbb{R}$, then $(a, b) =$

(1) $\left(2, -\frac{7}{2} \right)$

(2) $(2, -14)$

(3) $\left(-\frac{7}{2}, -14\right)$

(4) $(2, 7)$

65. The number of points in the interval $(0, 2)$ at which

$f(x) = |x - 0.5| + |x - 1| + \tan x$ is not differentiable is

(1) 1

(2) 2

(3) 3

(4) 4

66. If $x = \sec \theta - \cos \theta$, $y = \sec^{10} \theta - \cos^{10} \theta$ and

$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 = k(y^2 + 4)$, then $k =$

(1) $\frac{1}{100}$

(2) 1

(3) 10

(4) 100

67. If $y = e^{\sin^{-1}x}$, then $(1 - x^2)y_2 - xy_1 = 0$

- (1) 0
- (2) 1
- (3) y
- (4) $2y$

68. The semi vertical angle of a cone is 45° . If the height of cone is 20.025 cm, then the approximate value of its lateral surface area (in sq.cm) is

- (1) $401\sqrt{2}\pi$
- (2) $400\sqrt{2}\pi$
- (3) $402\sqrt{2}\pi$
- (4) $405\sqrt{2}\pi$

69. The radius of a sphere increase at the rate of 0.04 cm/sec. The rate of increase in the volume of that sphere with respect to its surface area, when its radius is 10 cm is

- (1) 16π
- (2) 25

(3) 20

(4) 5

70. If $f(x) = a \log|x| + bx^2 + x$ has extreme values at $x = -1$ and $x = 2$, then the ordered pair $(a, b) =$

(1) $(2, -1)$

(2) $\left(2, -\frac{1}{2}\right)$

(3) $(-1, 2)$

(4) $\left(-\frac{1}{2}, 2\right)$

71. If $f(x) = x^3 + px^2 + qx$ is defined on $[0, 2]$ such that

$f(0) = f(2)$ and $f'\left(1 + \frac{1}{\sqrt{3}}\right) = 0$, then $p^2 + q^2 =$

(1) 13

(2) 5

(3) $2 + \frac{1}{\sqrt{3}}$

(4) 1

72. If $x \neq \frac{-3}{\sqrt{2}}$, then $\int \frac{x^2}{2x^2 + 6\sqrt{2}x + 9} dx =$

(1) $\frac{1}{2\sqrt{2}} \left[(\sqrt{2}x + 3) - 6 \log |\sqrt{2}x + 3| - \frac{9}{\sqrt{2}x + 3} \right] + c$

(2) $\frac{1}{2\sqrt{2}} \left[(\sqrt{2}x + 3) - 6 \log |\sqrt{2}x + 3| + \frac{9}{\sqrt{2}x + 3} \right] + c$

(3) $\sqrt{2}x + 3 - 6 \log |\sqrt{2}x + 3| + c$

(4) $\log(2x^2 + 6\sqrt{2}x + 9) + c$

73. $\int \frac{\sqrt{1-x^2} \sin^{-1} x + x}{\sqrt{1-x^2}} dx =$

(1) $x \sin^{-1} x + \sqrt{1-x^2} + c$

(2) $\sin^{-1} x + \sqrt{1-x^2} + c$

(3) $x \sin^{-1} x + c$

(4) $\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + c$

74. If $\int x^3 e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$, then the sum of all the complex roots

of $f(x) = 1$ is

(1) $\frac{1}{2}$

(2) 3

(3) 1

(4) 2

75. $\int \sin^5 x \cos^5 x dx =$

(1) $\frac{\cos^6 x}{60} [6 \sin^4 x + 3 \sin^2 x + 1] + c$

(2) $-\frac{\sin^6 x}{60} [6 \cos^4 x - 3 \cos^2 x + 1] + c$

(3) $-\frac{\cos^6 x}{60} [6 \sin^4 x + 3 \sin^2 x + 1] + c$

(4) $\frac{\sin^6 x}{60} [6 \cos^4 x + 3 \cos^2 x + 1] + c$

76. $\lim_{x \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \frac{n^2}{(n+3)^3} + \dots + \frac{1}{125n} \right] =$

(1) $\frac{3}{8}$

(2) $\frac{15}{32}$

(3) $\frac{12}{25}$

(4) $\frac{35}{72}$

77. $\int_{e^{-1}}^{e^2} \left| \frac{\log x}{x} \right| dx =$

(1) $\frac{2}{5}$

(2) 2

(3) 5

(4) $\frac{5}{2}$

78. The area (in square units) bounded by $y = \tan^{-1}x$, $y = \cot^{-1}x$ and the Y-axis is

(1) $\log_e 4$

(2) $\log_e 2$

(3) $\log_e 3$

(4) $\log_e 5$

79. The order of the differential equation corresponding to the family of parabolas whose axes are along the X axis and whose foci are at the origin is

(1) 4

(2) 3

(3) 2

(4) 1

80. The general solution of the differential equation

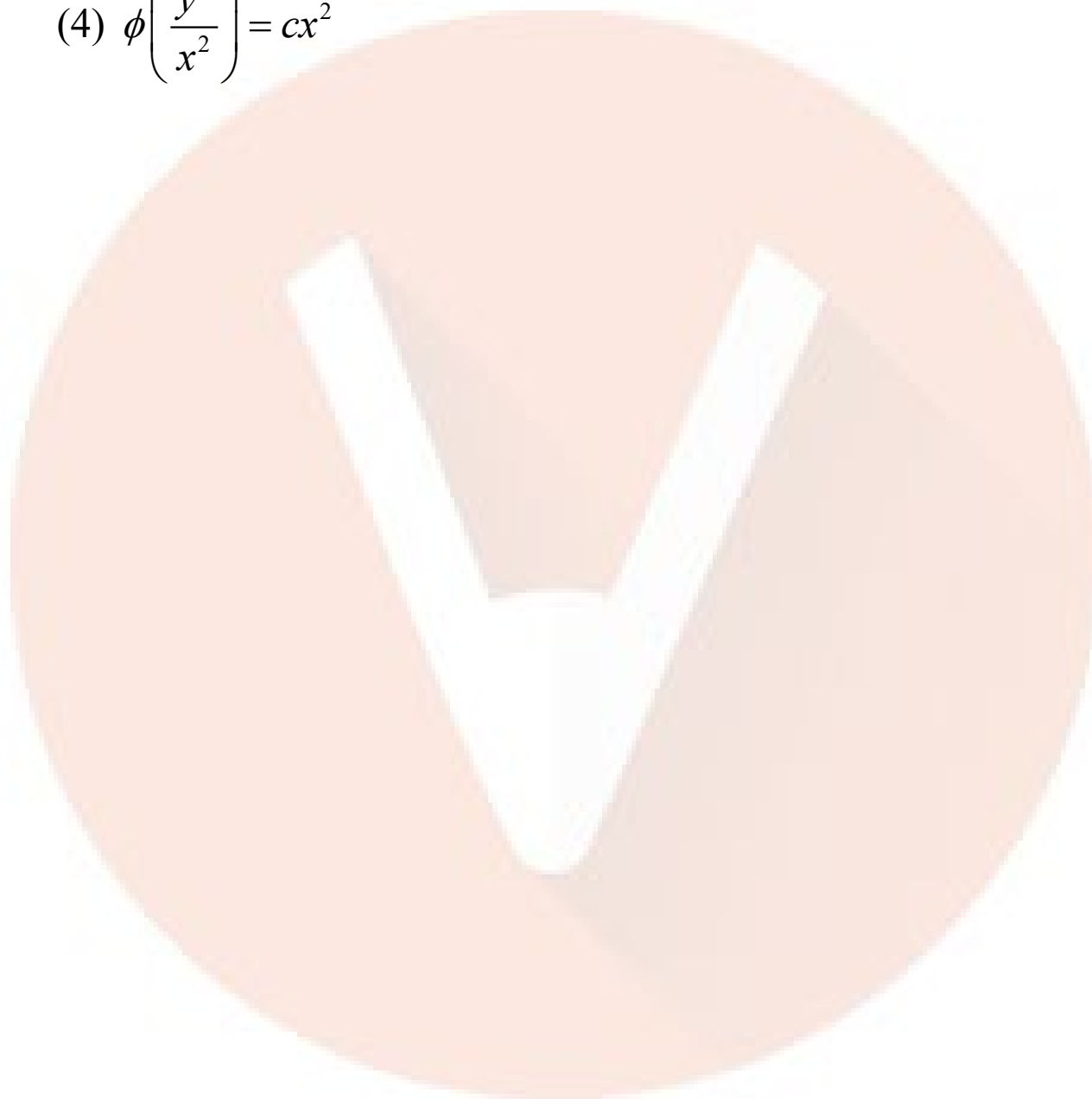
$$yy' = x \left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{x^2}{y^2}\right)}{\phi'\left(\frac{x^2}{y^2}\right)} \right], \text{ where } \phi \text{ is an arbitrary function, is}$$

(1) $x\phi\left(\frac{y^2}{x^2}\right) = cy$

(2) $x^2\phi\left(\frac{y^2}{x^2}\right) = c$

$$(3) \ x^2 \phi\left(\frac{y^2}{x^2}\right) = cy^2$$

$$(4) \ \phi\left(\frac{y^2}{x^2}\right) = cx^2$$



AP EAMCET-2019-Engineering-20 Apr-Shift-2
Answer Key

1	3	33	3	65	3	97	3	129	2
2	2	34	4	66	4	98	2	130	4
3	1	35	1	67	3	99	1	131	3
4	3	36	2	68	1	100	2	132	1
5	3	37	2	69	4	101	1	133	2
6	1	38	1	70	2	102	3	134	1
7	2	39	2	71	1	103	4	135	4
8	2	40	1	72	1	104	2	136	1
9	1	41	3	73	3	105	1	137	3
10	4	42	2	74	1	106	2	138	3
11	4	43	2	75	3	107	4	139	1
12	3	44	1	76	3	108	3	140	2
13	1	45	3	77	4	109	4	141	3
14	4	46	4	78	2	110	3	142	2
15	4	47	1	79	4	111	2	143	4
16	2	48	1	80	4	112	2	144	2
17	3	49	1	81	4	113	1	145	4

18	2	50	4	82	2	114	2	146	1
19	3	51	3	83	4	115	4	147	2
20	4	52	1	84	1	116	3	148	4
21	1	53	3	85	4	117	1	149	4
22	4	54	1	86	3	118	4	150	2
23	2	55	1	87	4	119	3	151	1
24	1	56	4	88	3	120	2	152	3
25	3	57	2	89	4	121	3	153	1
26	2	58	3	90	2	122	4	154	1
27	2	59	2	91	1	123	2	155	2
28	2	60	4	92	3	124	1	156	2
29	1	61	3	93	4	125	4	157	3
30	1	62	3	94	2	126	3	158	3
31	4	63	3	95	3	127	2	159	4
32	2	64	1	96	4	128	4	160	2

AP EAMCET-2019-Engineering-20 Apr-Shift-2
Solutions – Mathematics

1 Consider the function.

$$f(x) = \frac{3x+5}{7x-3}$$

Let $y = f(x)$

So,

$$y = \frac{3x+5}{7x-3}$$

Hence,

$$x = \frac{3y+5}{7y-3}$$

As $f(x)$ is a bijective function and

$$f^{-1}(x) = f(x).$$

Thus,

$$(fof)(x) = x$$

And,

$$(fofofof) = x$$

But,

$$(f \circ f \circ f) \neq x.$$

2. It is given that $f : A \rightarrow B$, $g : B \rightarrow C$ are the functions such that $g \circ f : A \rightarrow C$ is onto.

Then it is necessary that g is onto.

Thus, the correct option is (2).

3. Consider the given series,

$$\frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots n$$

The above expression can be simplified as,

$$\begin{aligned} \frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots n &= \left(1 - \frac{2}{5}\right) + \left(1 - \frac{4}{25}\right) + \left(1 - \frac{8}{125}\right) + \dots n \text{ terms} \\ &= n - \left[\frac{2}{5} + \left(\frac{2}{5}\right)^2 + \left(\frac{2}{5}\right)^3 + \dots n \text{ terms} \right] \\ &= n - \frac{\frac{2}{5} \left(1 - \left(\frac{2}{5}\right)^n\right)}{1 - \frac{2}{5}} \\ &= n + \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3} \end{aligned}$$

4. Consider the matrix,

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

It is solved as,

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 2-x & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{bmatrix}$$

The matrix is singular. Therefore,

$$\begin{vmatrix} 2-x & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{vmatrix} = 0$$

Apply operation $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} 5-x & 2 & 1 \\ 5-x & 3-x & 1 \\ 5-x & 2 & 2-x \end{vmatrix} = 0$$

$$(5-x) \begin{vmatrix} 1 & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{vmatrix} = 0$$

Apply operation $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$.

$$(5-x) \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1-x & 0 \\ 0 & 0 & 1-x \end{vmatrix} = 0$$

$$(5-x)(1-x)^2 = 0$$

$$x = 1, 5, 5$$

Thus, the sum of the required value of x is,

$$\text{Sum} = 1 + 1 + 5$$

$$= 7$$

5. Consider the matrix.

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$\begin{aligned} |A| &= \cos^2 \alpha + \sin^2 \alpha \\ &= 1 \end{aligned}$$

Now,

$$\begin{aligned} |\text{adj}A| &= |A|^{3-1} \\ &= 1^2 \\ &= 1 \end{aligned}$$

And,

$$\begin{aligned}\operatorname{adj}(\operatorname{adj} A) &= |A|^{3-2} \cdot A \\ &= 1 \cdot A \\ &= A\end{aligned}$$

Therefore,

$$\begin{aligned}(\operatorname{adj} A)^{-1} &= \frac{\operatorname{adj}(\operatorname{adj} A)}{|\operatorname{adj} A|} \\ &= \frac{A}{1} \\ &= A\end{aligned}$$

6. The given system of equation are,

$$x + 7ay + 2az = 0,$$

$$a + 6by + 2bz = 0$$

And,

$$x + 5cy + 2cz = 0$$

The system has non trivial solution. Therefore,

$$\begin{vmatrix} 1 & 7a & 2a \\ 1 & 6b & 2b \\ 1 & 5c & 2c \end{vmatrix} = 0$$

$$2bc - 4ac + 2ab =$$

$$2ac = bc + ab$$

$$b = \frac{2ac}{a+c}$$

Or,

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

Hence, a, b, c are in Harmonic progression.

7. It is given that,

$$|z - z_1| + |z - z_2| = k$$

And,

$$|z_1 - z_2| < k$$

Sum of the distance of the point z from z_1 and z_2 is constant (k) and more than the distance between points z_1 and z_2 . Thus, the locus of the point z is an ellipse.

8. It is given that,

$$\left\{ z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4} \right\}$$

Let the complex number be $z = x + iy$

Thus,

$$\begin{aligned} \frac{z-1}{z+1} &= \frac{(x-1) + iy}{(x+1) + iy} \times \frac{(x+iy) - iy}{(x+iy) - iy} \\ &= \frac{x^2 + y^2 - 1}{(x+1)^2 + y^2} + i \left(\frac{2y}{(x+1)^2 + y^2} \right) \end{aligned}$$

Therefore,

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

$$\left(\frac{2y}{x^2 + y^2 - 1} \right) = 1$$

$$x^2 + y^2 - 2y - 1 = 0$$

The complex plane represents a circle.

9. It is given that,

$$|z_1| = |z_2| = 1$$

And,

$$\operatorname{Re}(z_1 \bar{z}_2) = 0$$

Let,

$$\begin{aligned} z_1 &= a + ib \\ &= \cos \alpha + i \cos \beta \end{aligned}$$

And,

$$\begin{aligned} z_2 &= c + id \\ &= \cos \beta + i \sin \beta \end{aligned}$$

Thus,

$$\operatorname{Re}(z_1 \bar{z}_2) = 0$$

$$ac + bd = 0$$

$$\cos \alpha \cos \beta + \sin \alpha \sin \beta = 0$$

$$\cos(\alpha - \beta) = 0$$

Therefore,

$$\alpha - \beta = \frac{\pi}{2}, -\frac{\pi}{2}$$

The pair of complex number,

$$\begin{aligned} w_1 &= a + ic \\ &= \cos \alpha + i \cos \beta \end{aligned}$$

And,

$$\begin{aligned} w_2 &= b + id \\ &= \sin \alpha + i \sin \beta \end{aligned}$$

Hence,

$$\begin{aligned}\operatorname{Re}(w_1 \bar{w}_2) &= ab + cd \\ &= \cos \alpha \sin \alpha + \cos \beta \sin \beta \\ &= \frac{1}{2} 2 \sin(\alpha + \beta) \cos(\alpha - \beta) \\ &= 0\end{aligned}$$

This implies $\alpha - \beta = \frac{\pi}{2}, -\frac{\pi}{2}$.

10. Consider the expression.

$$(1+x)^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Substitute x for i in the above expression.

$$\begin{aligned}(1+i)^n &= \left[(a_0 - a_2 + a_4 - a_6 + \cdots) + \right. \\ &\quad \left. i(a_1 - a_3 + a_5 - a_7 + \cdots) \right] \\ \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]^n &= \left[(a_0 - a_2 + a_4 - a_6 + \cdots) + \right. \\ &\quad \left. i(a_1 - a_3 + a_5 - a_7 + \cdots) \right]\end{aligned}$$

Compare the real and imaginary parts.

$$2^{\frac{n}{2}} \cos \frac{n\pi}{4} = (a_0 - a_2 + a_4 - a_6 + \dots)$$

$$2^{\frac{n}{2}} \cos \frac{n\pi}{4} = k \cos \frac{n\pi}{4}$$

$$k = 2^{\frac{n}{2}}$$

11. Consider the equation.

$$\sqrt{x+2} - \sqrt{x-2} = \sqrt{4x-2}$$

Square both side,

$$x+2+x-2-2\sqrt{x^2-4} = 4x-2$$

$$1-x = \sqrt{x^2-4}$$

Now $x > 2$,

So,

$$1-x < 0$$

And,

$$\sqrt{x^2-4} > 0$$

Thus, there is no solution.

12. Consider the function.

$$y = \frac{x}{x^2 - 5x + 9}$$

The range of the function is determined as,

$$yx^2 - (5y + 1)x + 9y = 0.$$

The roots are real. Therefore,

$$(5y + 1)^2 - 36y^2 \geq 0$$

$$25y^2 + 10y + 1 - 36y^2 \geq 0$$

$$(11y + 1)(y - 1) \leq 0$$

$$y \in \left[-\frac{1}{11}, 1 \right]$$

13. Consider the cubic equations.

$$x^3 - ax^2 + bx - c = 0$$

The sum of the roots of the equation is,

$$\alpha + \beta + \gamma = a$$

And,

$$\alpha\beta + \beta\gamma + \gamma\alpha = b$$

The product of the roots is,

$$\alpha\beta\gamma = c$$

Therefore,

$$\begin{aligned}
\sum \alpha^2(\beta + \gamma) &= \sum \alpha^2(a - \alpha) \\
&= a \sum \alpha^2 - \sum \alpha^3 \\
&= \left[a \left[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \right] - \right. \\
&\quad \left. \left[(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha) \right] + \right. \\
&\quad \left. 3\alpha\beta\gamma \right] \\
&= a[a^2 - 2b] - [a(a^2 - 3b) + 3c]
\end{aligned}$$

Solve further,

$$\sum \alpha^2(\beta + \gamma) = ab - 3c$$

14. Consider the equation.

$$6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$$

Simplify the equation as follows,

$$\begin{aligned}
6(x^6 - 1) - 25x(x^4 - 1) + 31x^2(x^2 - 1) &= 0 \\
(x^2 - 1)[6(x^4 + x^2 + 1) - 25(x^2 + 1) + 31x^2] &= 0 \\
(x^2 - 1)[6x^4 - 25x^3 + 37x^2 - 25x + 6] &= 0
\end{aligned}$$

So,

$$x = \pm 1$$

Or,

$$6x^4 - 25x^3 + 37x^2 - 25x + 6 = 0$$

$$6(x^4 + 1) - 25x(x^2 + 1) + 37x^2 = 0$$

$$6\left(x + \frac{1}{x}\right)^2 - 25\left(x + \frac{1}{x}\right) + 37 = 0$$

$$3\left(x + \frac{1}{x}\right)\left[2\left(x + \frac{1}{x}\right) - 5\right] - 5\left[2\left(x + \frac{1}{x}\right) - 5\right] = 0$$

Solve further,

$$\left[2\left(x + \frac{1}{x}\right) - 5\right]\left[3\left(x + \frac{1}{x}\right) - 5\right] = 0$$

Thus,

$$2\left(x + \frac{1}{x}\right) - 5 = 0$$

Or,

$$3\left(x + \frac{1}{x}\right) - 5$$

Therefore,

$$2x^2 - 5x + 2 = 0$$

$$2x(x - 2) - 1(x - 2) = 0$$

$$(x - 2)(2x - 1) = 0$$

$$x = 2, \frac{1}{2}$$

And,

$$3x^2 - 5x + 3 = 0$$

It has no solution because $D < 0$.

The sum of all rational roots is,

$$\begin{aligned}\text{Sum} &= 1 - 1 + 2 + \frac{1}{2} \\ &= 2.5\end{aligned}$$

15. A number is divisible by 6 if it is divisible by both 2 and 3.

Thus, the last digit can be filled by 0, 2, 4 only.

The first place can be filled by 5 choices except 0 and second and third by 6 choices.

The fourth place is filled such that it must be of the form $3k$ or $3k + 1$ or $3k + 2$. Thus it can be filled by only two choices 0 or 3.

Hence the total numbers are,

$$\begin{aligned}\text{Number} &= 5 \times 6 \times 6 \times 2 \times 3 \\ &= 1080\end{aligned}$$

16. The number of ways to select 5 person out of 8 is 8C_5 .

Thus, the required number of ways is calculated as,

$$\begin{aligned}
 \text{Number of ways} &= {}^8C_5 \times 4! \times 2! \\
 &= \frac{8 \times 7 \times 6}{3 \times 2} \times 4 \times 3 \times 2 \times 2 \\
 &= 42 \times 64 \\
 &= 2688
 \end{aligned}$$

17. It is given that 15^k divides $47!$.

Find the exponent of 5 in $47!$.

$$\left[\frac{47}{5} \right] + \left[\frac{47}{5^2} \right] + \dots, \{ \text{where } x \text{ is greatest integer of } x \}$$

$$\begin{aligned}
 \text{exponent 5 of in } 47! &= 9 + 1 \\
 &= 10
 \end{aligned}$$

Thus, the value of k is 10.

18. Consider the equation.

$$(1 + x + x^2)^n = c_0 + c_1x + c_2x^2 + \dots$$

Substitute $\frac{1}{x}$ for x in the above equation.

$$\frac{(1 - x + x^2)^n}{x^{2n}} = \frac{c_0x^{2n} - c_1x^{2n-1} + c_2x^{2n-2} + c_3x^{2n-3} + \dots}{x^{2n}}$$

$$(1 - x + x^2)^n = c_0x^{2n} - c_1x^{2n-1} + c_2x^{2n-3} + \dots$$

Now,

$$\begin{aligned}
 c_0 x^{2n} + c_1 x^{2n-1} + c_2 x^{2n-3} + \dots &= \text{Coeff of } x^{2n+1} \text{ in } (1+x+x^2)^n (1-x+x^2)^n \\
 &= \text{Coeff of } x^{2n+1} \text{ in } \left((1+x^2)^2 - x^2 \right)^n \\
 &= \text{Coeff of } x^{2n+1} \text{ in } (1+x^4+x^2)^n \\
 &= 0
 \end{aligned}$$

Since the coefficient of x^k in the expansion of $(1+x^4+x^2)^n$ exist if k is even other it will be zero and $2n+1$ is odd

19. If x is so small that x^2 and higher power of x can be neglected, then,

$$(1+x)^n = 1 + nx$$

Thus,

$$\begin{aligned}
 \left(1 + \frac{3}{4}x\right)^{1/2} \left(1 - \frac{2}{3}x\right)^{-2} &= \left(1 + \frac{3}{4}x\right) \left(1 - \frac{2}{3}x\right) \\
 &= 1 + \frac{3}{8}x + \frac{4x}{3} \\
 &= \frac{24 + 41x}{24}
 \end{aligned}$$

20. Consider the equation.

$$\frac{x^4}{(x-a)(x-b)(x-c)} = P(x) + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

The above equation is solved as,

$$x^4 = \left[P(x)(x-a)(x-b)(x-c) + A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b) \right]$$

Substitute 0 for x in the above expression.

$$P(0)abc = [Abc + Bac + Cab]$$

$$P(0) = \frac{A}{a} + \frac{B}{b} + \frac{C}{c}$$

Now at $x = a$.

$$A = \frac{a^4}{(a-b)(a-c)}$$

At $x = b$

$$B = \frac{b^4}{(b-c)(b-a)}$$

At $x = c$

$$C = \frac{c^4}{(c-a)(c-b)}$$

Thus,

$$\begin{aligned}
 P(0) &= \frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-a)} \\
 &= \frac{a^3(c-b) + bc(c^2 - b^2) + a(b^3 - c^3)}{(a-b)(b-c)(c-a)} \\
 &= \frac{a^3 + bc(c+b) - a(b^2 + c^2 + bc)}{(a-b)(a-c)} \\
 &= \frac{a(a^2 - b^2) - c^2(a-b) - bc(a-b)}{(a-b)(a-c)}
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 P(0) &= \frac{(a^2 + ba) - c^2 - bc}{(a-c)} \\
 &= a + b + c
 \end{aligned}$$

Therefore,

$$P(0) + (a-b)(a-c)A = a + b + c + a^4$$

21. It is known that

$$\sin \alpha = p$$

$$\frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = p$$

The sum of the roots is,

$$\begin{aligned}\text{Sum} &= \tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \\ &= \frac{\tan^2 \frac{\alpha}{2} + 1}{\tan \frac{\alpha}{2}} \\ &= \frac{2}{p}\end{aligned}$$

The product of the roots is,

$$\begin{aligned}\text{Product} &= \tan \frac{\alpha}{2} \times \cot \frac{\alpha}{2} \\ &= 1\end{aligned}$$

Hence, the required quadratic equation is,

$$\begin{aligned}x^2 - \frac{2}{p}x + 1 &= 0 \\ px^2 - 2x + p &= 0\end{aligned}$$

22. Consider the expression.

$$\cos 66^\circ + \sin 84^\circ$$

It is solved as,

$$\begin{aligned}\cos 66^\circ + \sin 84^\circ &= 2\cos 36^\circ \cos 30^\circ \\ &= \sqrt{3}\cos 36^\circ \\ &= \sqrt{3}\left(\frac{\sqrt{5}+1}{4}\right)\end{aligned}$$

23. Consider the expression.

$$\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$$

It can be simplified as,

$$\frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = \tan \frac{\pi}{3}$$

$$\tan(3\theta) = \tan \frac{\pi}{3}$$

$$3\theta = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\theta = (3n+1)\frac{\pi}{9}, n \in \mathbb{Z}$$

24. Consider the equation.

$$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}(x)$$

The above equation can be rewritten as,

$$\sin^{-1}\left(\frac{3x}{5}\sqrt{1-\frac{16x^2}{25}} + \frac{4x}{5}\sqrt{1-\frac{9x^2}{25}}\right) = \sin^{-1}x$$

$$\frac{3x}{5}\sqrt{1-\frac{16x^2}{25}} + \frac{4x}{5}\sqrt{1-\frac{9x^2}{25}} = x$$

Thus,

$$x = 0$$

Or,

$$\frac{3\sqrt{25-16x^2}}{25} + \frac{4\sqrt{25-9x^2}}{25} = 1$$

$$9(25-16x^2) + 16(25-9x^2) + 24\sqrt{25-16x^2}\sqrt{25-9x^2} = 625$$

$$24\sqrt{25-16x^2}\sqrt{25-9x^2} = 625 - 625x^2$$

$$(25-16x^2)(25-9x^2) = 144x^4$$

Solve further,

$$625 - 625x^2 = 0$$

$$x = \pm 1$$

The sum of the values of x is,

$$\text{Sum} = 0 + 1 + -1$$

$$= 0$$

25. Consider the expression.

$$x = \log \left[\cot \left(\frac{\pi}{4} + \theta \right) \right]$$

$$e^x = \frac{\cot \pi/4 \cot \pi/6 - 1}{\cot \pi/4 + \cot \pi/6}$$

$$e^x = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$e^{-x} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

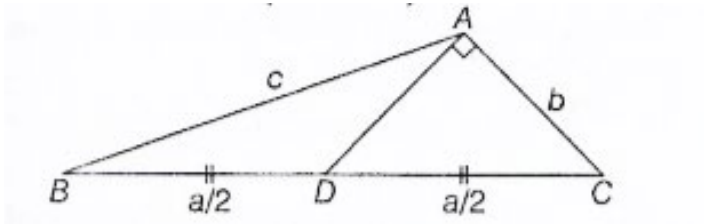
Now,

$$\begin{aligned} \sinh(x) &= \frac{e^x - e^{-x}}{2} \\ &= \frac{\frac{\sqrt{3} - 1}{\sqrt{3} + 1} - \frac{\sqrt{3} + 1}{\sqrt{3} - 1}}{2} \\ &= \frac{(3 + 1 - 2\sqrt{3}) - (3 + 1 + 2\sqrt{3})}{4} \\ &= -\frac{4\sqrt{3}}{4} \end{aligned}$$

Solve further,

$$\sinh(x) = -\sqrt{3}$$

26. Consider the figure shown below.



In $\triangle CAD$.

$$AD^2 = \frac{a^2}{4} - b^2$$

And from Apollonius theorem,

$$c^2 + b^2 = 2 \left(AD^2 + \frac{a^2}{4} \right)$$

Therefore,

$$c^2 + b^2 = 2 \left(\frac{a^2}{4} - b^2 + \frac{a^2}{4} \right)$$

$$c^2 + b^2 = a^2 - 2b^2$$

$$a^2 = 3b^2 + c^2$$

Thus,

$$\begin{aligned}
 3c \cos A \cos C + 2a^2 &= 3ca \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \left(\frac{a^2 + b^2 - c^2}{2ab} \right) + 2a^2 \\
 &= \frac{3}{4b^2} \left[(b^2 + c^2 - 3b^2 - c^2)(3b^2 + c^2 + b^2 - c^2) \right] + 2a^2 \\
 &= \frac{3}{4b^2} (-2b^2)(4b^2) + 2a^2 \\
 &= -6b^2 + 2a^2
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 3c \cos A \cos C + 2a^2 &= 2(a^2 - 3b^2) \\
 &= 2c^2
 \end{aligned}$$

27. It is known that,

$$\begin{aligned}
 \alpha^{-2} + \beta^{-2} + \gamma^{-2} &= \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2} \\
 &= \frac{a^2 + b^2 + c^2}{4\Delta^2} \\
 &= \frac{(a^2 + b^2 - c^2) + (b^2 + c^2 - a^2) + (a^2 + c^2 - b^2)}{4 \left(\frac{abc}{4R} \right)^2} \\
 &= \frac{2ab \cos C + 2bc \cos A + 2ac \cos B}{\frac{a^2 b^2 c^2}{4R^2}}
 \end{aligned}$$

Solve further,

$$\alpha^{-2} + \beta^{-2} + \gamma^{-2} = \frac{\cos C}{\left(\frac{abc}{4R}\right) \frac{c}{2R}} + \frac{\cos A}{\left(\frac{abc}{4R}\right) \frac{a}{2R}} + \frac{\cos B}{\left(\frac{abc}{4R}\right) \frac{b}{2R}}$$

$$= \frac{1}{\Delta} (\cot A + \cot B + \cot C)$$

28. Consider the expression.

$$(r_2 + r_3) \cot\left(\frac{B+C}{2}\right)$$

The above expression is solved as,

$$(r_2 + r_3) \cot\left(\frac{B+C}{2}\right) = \left(\frac{\Delta}{s-b} + \frac{\Delta}{s-c}\right) \tan \frac{A}{2}$$

$$= \frac{\Delta(s-c+s-b)}{(s-c)(s-b)} \frac{\Delta}{s(s-a)}$$

$$= 2s - b - c$$

$$= a$$

29. The given vectors are,

$$\overrightarrow{AB} = \bar{i} + 3\bar{j} + 4\bar{k}$$

$$\overrightarrow{AC} = 5\bar{i} + \bar{j} + 2\bar{k}$$

Therefore,

$$\begin{aligned}\overline{AG} &= \frac{(\bar{i} + 3\bar{j} + 4\bar{k}) + (5\bar{i} + \bar{j} + 2\bar{k})}{3} \\ &= 2\bar{i} + \frac{4}{3}\bar{j} + 2\bar{k}\end{aligned}$$

Thus,

$$\begin{aligned}|\overline{AG}| &= \sqrt{4 + \frac{16}{9} + 4} \\ &= \sqrt{\frac{88}{9}} \\ &= \frac{2}{3}\sqrt{22}\end{aligned}$$

30. Let point P be the intersection of lines

$$\bar{r} = (\bar{i} + 2\bar{j} - \bar{k}) + \lambda(2\bar{i} + 3\bar{j} + 4\bar{k}) \quad \text{and}$$

$$\bar{r} = (-\bar{i} - 3\bar{j} + 7\bar{k}) + \mu(\bar{i} + 2\bar{j} - \bar{k})$$

Thus,

$$2\lambda + 1 = \mu - 1 \quad \dots\dots (I)$$

$$3\lambda + 2 = -3 + 2\mu \quad \dots\dots (II)$$

And,

$$4\lambda - 1 = 7 - \mu \quad \dots\dots (III)$$

On solving the above equations,

$$\lambda = 1$$

$$\mu = 4$$

Thus the required point of intersection is,

$$\bar{P} = (3\bar{i} + 5\bar{j} + 3\bar{k})$$

31. The position vector of A is $\bar{i} + \bar{j} + \bar{k}$, B is $4\bar{i} - \bar{j} + 2\bar{k}$, C is $5\bar{i} + \bar{j}$ and D is $7\bar{i} + 2\bar{j} + 3\bar{k}$.

Thus,

$$\overline{AB} = 3\bar{i} - 2\bar{j} + \bar{k}$$

And,

$$\overline{CD} = 2\bar{i} + \bar{j} + 3\bar{k}$$

Therefore, the projection of $\overline{AB}$ on $\overline{CD}$ is,

$$\begin{aligned}\text{Projection} &= \frac{|6 - 2 + 3|}{\sqrt{4 + 1 + 9}} \\ &= \frac{7}{\sqrt{14}} \\ &= \sqrt{\frac{7}{2}}\end{aligned}$$

32. It is given that,

$$\frac{1}{2}|\vec{a} - \vec{b}| = \sin \lambda \theta$$

Square both sides.

$$|\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos \theta = 4\sin^2(\lambda \theta)$$

$$1 + 1 - 2\cos \theta = 4\sin^2(\lambda \theta)$$

$$2(1 - \cos \theta) = 4\sin^2(\lambda \theta)$$

$$4\sin^2\left(\frac{\theta}{2}\right) = 4\sin^2(\lambda \theta)$$

Solve further,

$$\lambda = \frac{1}{2}$$

Therefore,

$$\begin{aligned} 4\lambda^2 &= 4\left(\frac{1}{2}\right)^2 \\ &= 1 \end{aligned}$$

33. Consider the give vectors.

$$\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$$

$$\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$$

And,

$$\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$$

Thus,

$$\bar{a} + \bar{b} + \bar{c} = 6\bar{i} - 2\bar{j} + 3\bar{k}$$

$$\bar{a} - \bar{b} + \bar{c} = 4\bar{i} - 6\bar{j} + 7\bar{k}$$

And,

$$\bar{a} + \bar{b} - \bar{c} = \bar{k}$$

Hence, the required volume of the parallelepiped is calculated

as,

$$\begin{aligned} V &= \left| \begin{vmatrix} 6 & -2 & 3 \\ 4 & -6 & 7 \\ 0 & 0 & 1 \end{vmatrix} \right| \\ &= |1(-36 + 8)| \\ &= |-28| \\ &= 28 \end{aligned}$$

34. Consider the expression.

$$\bar{a} \times (\bar{a} \times \bar{c}) + \bar{b} = \bar{0}$$

It is solved as,

$$(\bar{a} \cdot \bar{c})\bar{a} - (\bar{a} \cdot \bar{a})\bar{c} = -\bar{b}$$

$$\text{Let } \bar{a} \cdot \bar{c} = x$$

Thus,

$$x\bar{a} - \bar{c} = -\bar{b}$$

Square both sides

$$x^2 |\bar{a}|^2 + |\bar{c}|^2 - 1x |\bar{a} \cdot \bar{c}| = |\bar{b}|^2$$

$$x^2 + 4 - 2x^2 = 1$$

$$x^2 = 3$$

$$(\bar{a} \cdot \bar{c})^2 = 3$$

35. According to the given data,

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-8	64
3	-7	49
5	-5	25
7	-3	9
11	1	1
13	3	9
17	7	49
22	12	144
$\sum x_i = 80$		$\sum (x_i - \bar{x})^2 = 350$

Thus,

$$\begin{aligned}\bar{x} &= \frac{80}{8} \\ &= 10\end{aligned}$$

Variance is calculated as,

$$\begin{aligned}\text{Variance} &= \frac{\sum (x_i - \bar{x})^2}{8} \\ &= \frac{350}{8} \\ &= 43.75\end{aligned}$$

36. The expression for the coefficient of variance is given by,

$$CV = \frac{\sigma}{\mu} \times 100$$

Therefore,

$$\begin{aligned}60 &= \frac{21}{\mu} \times 100 \\ \mu &= \frac{21 \times 10}{6} \\ &= 35\end{aligned}$$

The new mean on addition of 15 to every observation is,

$$\begin{aligned}\mu' &= 35 + 15 \\ &= 50\end{aligned}$$

But,

$$\sum |\mu' - x'_i| = \sum |\mu - x_i|$$

Now,

$$21 = \sqrt{\frac{\sum (\mu - x_i)^2}{n}}$$
$$21^2 = \frac{\sum (\mu - x_i)^2}{n}$$

$$\sum (\mu - x_i)^2 = 21^2 \times n$$

Thus,

$$\begin{aligned}\sigma' &= \sqrt{\frac{\sum (\mu' - x'_i)^2}{n}} \\ &= \sqrt{\frac{21^2 \times n}{n}} \\ &= 21\end{aligned}$$

Therefore, the new coefficient of variance is,

$$\begin{aligned}CV_{\text{New}} &= \frac{21}{50} \times 100 \\ &= 42\end{aligned}$$

37. The required probability is calculated as,

$$\begin{aligned}
 P &= \frac{9+8+7+6+5+4+3+2+1}{10 \times 10} \\
 &= \frac{45}{100} \\
 &= \frac{9}{20}
 \end{aligned}$$

38. Let E_1 be the event of selecting word LONDON and E_2 be the event of selecting the word CLIFTON.

Thus, the required probability is calculated as,

$$\begin{aligned}
 P\left(\frac{E_1}{A}\right) &= \frac{P(E_1) \cdot P\left(\frac{A}{E_1}\right)}{P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right)} \\
 &= \frac{\frac{1}{2} \cdot \frac{2}{5}}{\left(\frac{1}{2} \cdot \frac{2}{5}\right) + \left(\frac{1}{2} \cdot \frac{1}{6}\right)} \\
 &= \frac{2 \times 6}{12 + 5} \\
 &= \frac{12}{17}
 \end{aligned}$$

39. The given probability of the function is,

$$P(X = k) = Ck^2$$

Therefore,

$$\sigma^2 = E(X^2) - \mu^2$$

$$\sigma^2 + \mu^2 = E(X^2)$$

$$\begin{aligned}\sigma^2 + \mu^2 &= \sum (x_i^2) P(x_i) \\ &= 0(c(0)^2) + 1(c(1)^2) + 4(c(2)^2) + 9(c(3)^2) + 16(c(4)^2) \\ &= c + 16c + 81c + 256c\end{aligned}$$

Now,

$$\begin{aligned}\sum P(x_i) &= 1 \\ c[0^2 + 1^2 + 2^2 + 3^2 + 4^2] &= 1 \\ c &= \frac{1}{30}\end{aligned}$$

Solve further,

$$\begin{aligned}\sigma^2 + \mu^2 &= 354c \\ &= \frac{354}{30} \\ &= 11.8\end{aligned}$$

40. According to the given data,

$$\lambda = \frac{200}{250}$$
$$= \frac{4}{5}$$

Now,

$$P(X = x) = \frac{e^{-\lambda} (\lambda^x)}{x!}$$

$$P(X = 0) = e^{-\frac{4}{5}}$$

Therefore, the required probability is calculated as,

$$\begin{aligned} (P(X = 0))^5 &= \left(e^{-\frac{4}{5}} \right)^5 \\ &= e^{-4} \end{aligned}$$

41. Consider the given equation.

$$\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4$$

Sum of the distance of point $P(x, y)$ with respect to points $A(2, 0)$ and $B(-2, 0)$ is equal to the distance between the points $A(2, 0)$ and $B(-2, 0)$.

$$AP + BP = AB$$

Therefore, the locus of point $P(x, y)$ is a line segment.

42. Consider the given equation.

$$2x^2 + 2y^2 - 8x - 12y + 18 = 0$$

It can be simplified as,

$$2(x-2)^2 + 2(y-3)^2 = 8$$

$$(x-2)^2 + (y-3)^2 = 4$$

After shifting the origin to $(2,3)$, the transformed equation becomes $x^2 + y^2 = 4$ and after rotation of axes through an angle of 45° about the point, the transformed equation is $x^2 + y^2 = 4$.

43. Let a line cuts the coordinate axis at A and B. So,

$$\frac{x}{a} + \frac{y}{b} = 1$$

Now, $A(a,0)$, $B(0,b)$ and the line passes through the point $P(4,2)$.

Thus,

$$\frac{4}{a} + \frac{2}{b} = 1$$

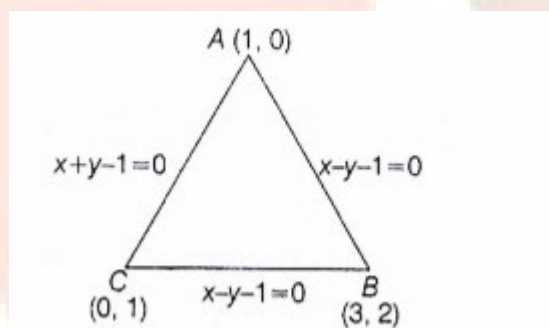
The coordinate of centre of the circumcentre of ΔOAB is midpoint of hypotenuse of the right angled ΔOAB and its midpoint is AB . So, the circumcentre is $\left(\frac{a}{2}, \frac{b}{2}\right)$.

Therefore,

$$\frac{2}{x} + \frac{1}{y} = 1$$

$$2x^{-1} + y^{-1} = 1$$

44. Consider the figure shown below.



The points of intersection of the lines are $A(1, 0)$, $B(3, 2)$ and $C(0, 1)$.

The centroid of ΔABC is calculated as,

$$\text{Centroid} = \left(\frac{1+3+0}{3}, \frac{0+2+1}{3} \right)$$

$$= \left(\frac{4}{3}, 1 \right)$$

45. The equation of perpendicular bisector of the line joining the points $A(-2, -1)$ and $B(6, -1)$ is,

$$x = 2$$

The equation of perpendicular bisector of the line joining the points $B(6, -1)$ and $C(2, 5)$ is,

$$y - 2 = \frac{4}{6}(x - 4)$$

$$2x - 3y = 2$$

The point of intersection of perpendicular bisector of the circumcentre of $\triangle ABC$.

$$C\left(2, \frac{2}{3}\right)$$

Thus, the required quadratic equation is,

$$x^2 - \left(2 + \frac{2}{3}\right)x + 2\left(\frac{2}{3}\right) = 0$$
$$3x^2 - 8x + 4 = 0$$

46. The point of intersection of the lines $x + y + 1 = 0$ and

$$2x - 3y + 4 = 0 \text{ is } A\left(-\frac{7}{5}, \frac{2}{5}\right).$$

If the coordinate of the image of point $P(-2,0)$ with respect to line $x + y + 1 = 0$ is (h,k) , then

$$\frac{h+2}{1} = \frac{k-0}{1} = 2 \times \frac{-2+1}{2} = 1$$

So,

$$h = -1$$

$$k = 1$$

Thus,

$$(h,k) = (-1,1)$$

The equation of required bisector of angle is line joining the points $A\left(-\frac{7}{5}, \frac{2}{5}\right)$ and $B(-1,1)$ is

$$y - 1 = \frac{\frac{3}{5}}{\frac{5}{5}}(x + 1)$$

$$2y - 2 = 3x + 3$$

$$3x - 2y + 5 = 0$$

47. The given equation of lines are,

$$3x^2 + 2hxy - 3y^2 = 0$$

And,

$$3x^2 + 2hxy - 3y^2 + 2x - 4y + c = 0$$

The point of intersection of the lines given by

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

Thus, the point of intersection of the second line is,

$$A(x, y) = A\left(\frac{2h - 3}{9 + h^2}, -\frac{h + 6}{9 + h^2}\right)$$

The equation of diagonal of square passes through origin is,

$$y = \frac{h + 6}{3 - 2h}x.$$

The equation of diagonal of square that does passes through the origin is

$$2x - 4y + c = 0$$

The diagonal are perpendicular if,

$$\frac{1}{2} \left(\frac{h + 6}{3 - 2h} \right) = -1$$

$$h + 6 = 4h - 6$$

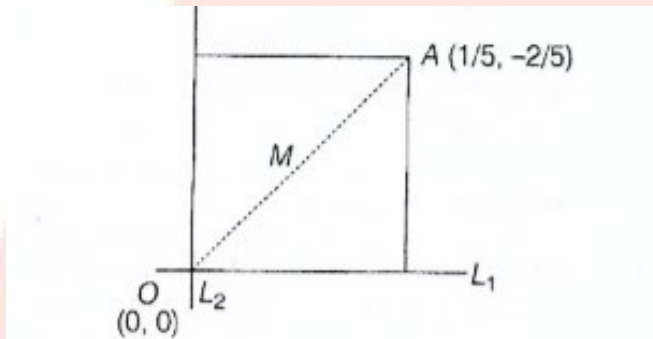
$$3h = 12$$

$$h = 4$$

Thus,

$$\begin{aligned}\text{Point } A &= \left(\frac{8-3}{9+16}, -\frac{10}{9+16} \right) \\ &= \left(\frac{1}{5}, -\frac{2}{5} \right)\end{aligned}$$

Consider the figure shown below.



The midpoint of line OA is,

$$M = \left(\frac{1}{10}, -\frac{1}{5} \right)$$

Therefore,

$$\frac{2}{10} + \frac{4}{5} + c = 0$$

$$c = -1$$

Therefore, $(h, c) = (4, -1)$.

48. Consider the equation of curve,

$$x^2 + xy + y^2 + x + 3y + 1 = 0$$

The equation of line joining the origin to the point of intersection of curve and the line is calculated as,

$$x^2 + xy + y^2 + (x + 3y)\left(\frac{x + y}{-2}\right) + \left(\frac{x + y}{-2}\right)^2 = 0$$

$$x^2 + y^2 + xy - \frac{1}{2}(x^2 + 4xy + 3y^2) + \frac{1}{4}(x^2 + 2xy + y^2) = 0$$

$$3x^2 - 2xy - y^2 = 0$$

The equation of the bisector of the angle between the pair of straight line is,

$$\frac{x^2 - y^2}{3 + 1} = \frac{xy}{-1}$$

$$x^2 + 4xy - y^2 = 0$$

49. Consider the equation of circle.

$$x^2 + y^2 - 2x - 4y - 20 = 0$$

The equation of tangent at the point B(1, 7) on the circle is,

$$x + 7y - (x + 1) - 2(y + 7) - 20 = 0$$

$$5y = 35$$

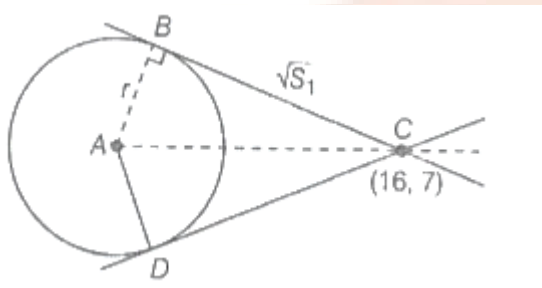
$$y = 7$$

The equation of tangent at the point D(4, -2) on the circle is,

$$4x - 2y - (x + 4) - 2(y - 2) - 20 = 0$$

$$3x - 4y = 20$$

Consider the figure shown below.



The point of intersection of tangents is $C(16, 7)$.

Thus, the area of quadrilateral ABCD is,

$$\begin{aligned} \text{Area} &= 2 \times \frac{1}{2} r \sqrt{S_1} \\ &= 2 \times \frac{1}{2} \times (5) \sqrt{256 + 49 - 32 - 28 - 20} \\ &= 5 \times \sqrt{225} \\ &= 75 \end{aligned}$$

50. The all conjugate of lines $5x + 7y - 78 = 0$ with respect to the circle $x^2 + y^2 + 6x + 8y - 96 = 0$ passes through the pole of the given line with respect to given circle.

Let required pole be $P(x_1, y_1)$ with respect to the circle $T = 0$.

$$xx_1 + yy_1 + 3(x + x_1) + 4(y + y_1) - 96 = 0$$
$$(x_1 + 3)x + (y_1 + 4)y + (3x_1 + 4y_1 - 96) = 0$$

The above line represent the line $5x + 7y - 78 = 0$ only.

Hence,

$$\frac{x_1 + 3}{5} = \frac{y_1 + 4}{7} = \frac{3x_1 + 4y_1 - 96}{-78} = k$$

Therefore,

$$x_1 = 5k - 3$$

$$y_1 = 7k - 4$$

And,

$$3x_1 + 4y_1 = 96 - 78k$$

Thus,

$$3(5k - 3) + 4(7k - 4) = 96 - 78k$$

$$121k = 121$$

$$k = 1$$

So,

$$x_1 = 2$$

$$y_1 = 3$$

Therefore, the required point of concurrence of all conjugate lines of the given line with respect to the given circle is $(2, 3)$.

51. Consider the given equation of circles.

$$S_1 : x^2 + y^2 + 4x - 6y - 12 = 0$$

Its centre is $(-2, 3)$ and radius is $r_1 = 5$

And,

$$S_2 : x^2 + y^2 - 8x + 10y + 5 = 0$$

Its centre is $(4, -5)$ and radius is $r_2 = 6$.

So,

$$\begin{aligned} r_1 + r_2 &= 5 + 6 \\ &= 11 \end{aligned}$$

And,

$$\begin{aligned} C_1 C_2 &= \sqrt{(4 + 2)^2 + (-5 - 3)^2} \\ &= \sqrt{36 + 64} \\ &= \sqrt{100} \\ &= 10 \end{aligned}$$

Since,

$$r_1 + r_2 > C_1 C_2$$

Therefore, there are only two common tangents to circle S_1 and S_2 .

52. Consider the given circles.

$$S_1 : (x-1)^2 + (y-3)^2 = r^2$$

And,

$$S_2 : (x-4)^2 + (y+1)^2 = 3^2$$

If the circle intersect at two distinct points, then

$$|r_1 - r_2| < C_1 C_2 < (r_1 + r_2)$$

$$|r - 3| < \sqrt{(4-1)^2 + (-1-3)^2} < (r + 3)$$

$$|r - 3| < 5 < (r + 3)$$

So,

$$r + 3 > 5$$

$$r > 2$$

And,

$$|r - 3| < 5$$

$$r - 3 \in (-5, 5)$$

$$r \in (-2, 8)$$

Thus, $2 < r < 8$.

53. Consider the circles.

$$S_1 = x^2 + y^2 + 6x - 2y + k = 0$$

And,

$$S_2 = x^2 + y^2 + 2x - 6y - 15 = 0$$

The circle S_1 bisects the circumference of the circle S_2 , then the common circles S_1 and S_2 will pass through the circle $S_2(-1,3)$.

The equation of common chord of the circle S_1 and S_2 is

$$4x + 4y + (k + 15) = 0$$

Since the chord passes through point $(-1,3)$. Thus,

$$-4 + 12 + k + 15 = 0$$

$$k = -23$$

54. Consider the given circles.

$$S_1 : x^2 + y^2 + 4y - 7 = 0$$

$$S_2 : 2x^2 + 2y^2 + 3x + 5y - 9 = 0$$

And,

$$S_3 : x^2 + y^2 + y = 0$$

The radical axis of circle S_1 and S_2 is $x - y - 1 = 0$.

The radical axis of circle S_2 and S_3 is $x + y - 3 = 0$.

Thus, the coordinate of radical centre is $(2,1)$ and the radius is,

$$\begin{aligned} r &= \sqrt{4 + 1 + 8 - 7} \\ &= \sqrt{6} \end{aligned}$$

Hence, the equation of the required circle is,

$$(x-2)^2 + (y-1)^2 = 6$$

$$x^2 + y^2 - 4x - 2y - 1 = 0$$

55. Consider the given equation of parabola.

$$2y^2 + 5x - 6y + 1 = 0$$

It can be simplified as,

$$2(y^2 - 3y) + 5x + 1 = 0$$

$$2\left(y^2 - 3y + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2\right) + 5x + 1 = 0$$

$$2\left(y - \left(\frac{3}{2}\right)\right)^2 - \frac{9}{2} + 5x + 1 = 0$$

$$\left(y - \frac{3}{2}\right)^2 = 4\left(-\frac{5}{8}\right)\left(x - \frac{7}{10}\right)$$

The coordinate of the vertex of parabola is $\left(\frac{7}{10}, \frac{3}{2}\right)$ and the

coordinate of the focus of the parabola is $\left(\frac{3}{40}, \frac{3}{2}\right)$

56. Consider the equation of parabola

$$y^2 = 9x$$

The equation of normal chord at a point t on the parabola is

$$tx + y = 2\left(\frac{9}{4}\right)t + \frac{9}{4}t^3$$

The chord subtends right angle at vertex of parabola $V(0,0)$.

Thus,

$$y^2 - 9x \left(\frac{tx + y}{\frac{9}{2}t + \frac{9}{4}t^3} \right) = 0$$

The sum of the coefficient of x^2 and y^2 terms should be zero.

Hence,

$$1 - \frac{9t}{\frac{9}{2}t + \frac{9}{4}t^3} = 0$$

$$1 - \frac{1}{\frac{1}{2} + \frac{t^2}{4}} = 0$$

$$\frac{1}{2} + \frac{t^2}{4} = 1$$

$$t^2 = 2$$

Therefore,

$$t = \pm\sqrt{2}$$

57. Let the equation of ellipse be,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The length of the latus rectum is,

$$\frac{2b^2}{a} = 4$$

The distance between the foci is,

$$2ae = 4\sqrt{2}$$

$$a\sqrt{1 - \frac{b^2}{a^2}} = 2\sqrt{2}$$

$$a^2 - b^2 = 8$$

$$a^2 - 2a - 8 = 0$$

Solve further,

$$(a - 4)(a + 2) = 0$$

$$a = 4, -2$$

Since, $a > 0$, $a = 4$.

Hence,

$$b^2 = 8$$

Therefore, the required ellipse is,

$$\frac{x^2}{16} + \frac{y^2}{8} = 1$$
$$x^2 + 2y^2 = 16$$

58. Consider the equation of ellipse

$$\frac{x^2}{4} + \frac{y^2}{1} = 1$$

So,

$$c^2 = 4(4) + 1$$

$$c^2 = 16 + 1$$

$$c^2 = 17$$

$$c = \pm\sqrt{17}$$

Now consider the equation.

$$x^3 - x^2 - 17x + 17 = 0$$

$$x^2(x-1) - 17(x-1) = 0$$

$$(x-1)(x^2 - 17) = 0$$

$$x = \pm\sqrt{17}, 1$$

Thus, option (3) is correct.

59. Consider the hyperbola.

$$9x^2 - 16y^2 = 144$$

Its eccentricity is e_1

And,

$$9x^2 - 16y^2 = -144$$

Its eccentricity is e_2 and is a conjugate hyperbola.

Therefore,

$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

$$\frac{e_1^2 + e_2^2}{e_1^2 e_2^2} = 1$$

$$\frac{e_1^2 e_2^2}{e_1^2 + e_2^2} = 1$$

60. Consider the points.

$$A = (2, 4, -1)$$

$$B = (3, 6, -1)$$

And,

$$C = (4, 5, 1)$$

$$\text{Let } D = (x, y, z)$$

The midpoint of AC is equal to the midpoint of BD .

$$\left(\frac{2+4}{2}, \frac{4+5}{2}, \frac{-1+1}{2}\right) = \left(\frac{3+x}{2}, \frac{6+y}{2}, \frac{-1+z}{2}\right)$$
$$\left(3, \frac{9}{2}, 0\right) = \left(\frac{3+x}{2}, \frac{6+y}{2}, \frac{-1+z}{2}\right)$$

Therefore,

$$x = 3$$

$$y = 3$$

$$z = 1$$

Hence, the fourth vertex is $D(3,3,1)$.

61. Consider the points.

$$A = (2, 3, -1)$$

$$B = (3, 5, -3)$$

The DR's of AB are,

$$\begin{aligned} \text{DR's } AB &= (3-2, 5-3, -3+1) \\ &= (1, 2, -2) \end{aligned}$$

And,

$$C = (1, 2, 3)$$

$$D = (3, y, 7)$$

$$\begin{aligned}\text{DR's } CD &= (3-1, y-2, 7-3) \\ &= (2, y-2, 4)\end{aligned}$$

It is given that $AB \perp CD$

Hence,

$$\begin{aligned}a_1a_2 + b_1b_2 + c_1c_2 &= 0 \\ (1)(2) + 2(y-2) + (-2)(4) &= 0 \\ 2y - 10 &= 0 \\ y &= 5\end{aligned}$$

62. It is given that

$$\text{X-intercept}(a) = 2$$

$$\text{Y-intercept}(b) = 3$$

$$\text{Z-intercept}(c) = 4$$

Therefore, the equation of the plane is,

$$\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$$

$$6x + 4y + 3z = 12$$

The given points are $A(-1, 6, 2)$, $B(1, 2, 3)$ and $C(-2, 3, 4)$.

Thus,

$$\begin{aligned}\text{DR's of } BC &= (-2-1, 3-2, 4-3) \\ &= (-3, 1, 1)\end{aligned}$$

Therefore, the equation of plane passing through $A(-1, 6, 2)$ and

DR's $(-3, 1, 1)$ is,

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$-3(x + 1) + 1(y - 6) + 1(z - 2) = 0$$

$$-3x + y + z - 11 = 0$$

$$-3x + y + z = -11$$

Hence, the angle between the planes is,

$$\begin{aligned}\cos \theta &= \frac{|(6)(-3) + (4)(1) + (3)(1)|}{\sqrt{36 + 16 + 9} \sqrt{9 + 1 + 1}} \\&= \frac{|-18 + 4 + 3|}{\sqrt{61} \sqrt{11}} \\&= \frac{\sqrt{11}^2}{\sqrt{61} \sqrt{11}} \\&= \sqrt{\frac{11}{61}}\end{aligned}$$

Solve further,

$$\theta = \cos^{-1} \left(\sqrt{\frac{11}{61}} \right)$$

63. Consider the equation.

$$\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$$

And,

$$\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$$

Simplify the first equation as,

$$\begin{aligned}\alpha &= \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x} \\&= \lim_{x \rightarrow 0} \frac{x(2^x - 1)}{2 \sin^2 \frac{x}{2}} \\&= \lim_{x \rightarrow 0} \frac{(2^x - 1)}{2 \sin^2 \frac{x}{2}} \\&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(\frac{2^x - 1}{x} \right)}{\frac{\sin^2 \frac{x}{2}}{x^2}}\end{aligned}$$

Solve further,

$$\alpha = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(\frac{2^x - 1}{x} \right)}{\left(\frac{\sin \frac{1}{2} x}{\frac{x}{2}} \right)^2}$$

$$= \frac{1}{2} \left[\lim_{x \rightarrow 0} \frac{2^x - 1}{x} \times \frac{1}{\lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{2} x}{\frac{x}{2}} \right)^2 \times \frac{1}{4}} \right]$$

$$= \frac{1}{2} \times \log 2 \times \frac{1}{1 \times \frac{1}{4}}$$

$$= 2 \log 2$$

Also,

$$\begin{aligned}\beta &= \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}} \times \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \\&= \lim_{x \rightarrow 0} \frac{x(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{1+x^2 - (1-x^2)} \\&= \lim_{x \rightarrow 0} \frac{x(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{2x^2} \\&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{x}\end{aligned}$$

Solve further,

$$\beta = \frac{1}{2} \times \log 2 \times 2$$

$$\beta = \log 2$$

$$2\beta = 2 \log 2$$

$$2\beta = \alpha$$

64. Consider the piecewise function,

$$f(x) = \begin{cases} \frac{x - |x|}{x} & , \text{ when } x < 0 \\ 5x^2 + a & , \text{ when } 0 \leq x \leq 1 \\ b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right) & , \text{ when } 1 < x < 3 \\ -14 & , \text{ when } x \geq 3 \end{cases}$$

The function is continuous at $x = 0$ and $x = 3$.

LHL at $x = 0$ is equal to RHL at $x = 0$.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} \frac{x - |x|}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

$$\lim_{x \rightarrow 0^-} \frac{x - (-x)}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

$$\lim_{x \rightarrow 0^-} \frac{2x}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

Solve further,

$$2 = a$$

$$a = 2$$

The function is continuous at $x = 3$.

LHL at $x = 3$ is equal to RHL at $x = 3$.

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right) = \lim_{x \rightarrow 3^+} -14$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{(x+1)(x-1)}{(x-1)(x-2)} \right) = \lim_{x \rightarrow 3^+} -14$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{(x+1)}{(x-2)} \right) = \lim_{x \rightarrow 3^+} -14$$

Solve further,

$$b \frac{(3+1)}{(3-2)} = -14$$

$$4b = -14$$

$$b = \frac{-14}{4}$$

$$b = -\frac{7}{2}$$

Therefore,

$$(a, b) = \left(2, -\frac{7}{2} \right)$$

65. Consider the given equation.

$$f(x) = |x - 0.5| + |x - 1| + \tan x$$

Clearly $f(x)$ is not differentiable at $x = 0.5$, 1 and $\frac{\pi}{2}$ for $x \in (0, 2)$.

Therefore, the number of non differentiable points of $f(x)$ are 3.

66. It is given that,

$$x = \sec \theta - \cos \theta$$

$$y = \sec^{10} \theta - \cos^{10} \theta$$

And,

$$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 = k(y^2 + 4)$$

Therefore,

$$\frac{dy}{d\theta} = 10(\sec^{10} \theta + \cos^{10} \theta) \tan \theta$$

And,

$$\frac{dx}{d\theta} = (\sec \theta + \cos \theta) \tan \theta$$

Hence,

$$\begin{aligned}\frac{dy}{dx} &= 10 \frac{\sec^{10} \theta + \cos^{10} \theta}{\sec \theta + \cos \theta} \\ \left(\frac{dy}{dx}\right)^2 &= 100 \left(\frac{\sec^{10} \theta + \cos^{10} \theta}{\sec \theta + \cos \theta}\right)^2 \\ &= 100 \left(\frac{\sec^{20} \theta + \cos^{20} \theta + 2}{\sec^2 \theta + \cos^2 \theta + 2}\right) \\ &= 100 \left(\frac{(\sec^{10} \theta - \cos^{10} \theta)^2 + 4}{(\sec \theta - \cos \theta)^2 + 4}\right)\end{aligned}$$

Solve further,

$$\begin{aligned}\left(\frac{dy}{dx}\right)^2 &= 100 \left(\frac{y^2 + 4}{x^2 + 4}\right) \\ (x^2 + 4) \left(\frac{dy}{dx}\right)^2 &= 100(y^2 + 4) \\ &= k(y^2 + 4)\end{aligned}$$

Therefore,

$$k = 100$$

67. Consider the equation.

$$y = e^{\sin^{-1} x}$$

Differentiate with respect to x .

$$y_1 = e^{\sin^{-1}x} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$y_1 \sqrt{1-x^2} = e^{\sin^{-1}x}$$

$$y_1 \sqrt{1-x^2} = y$$

Differentiate with respect to x

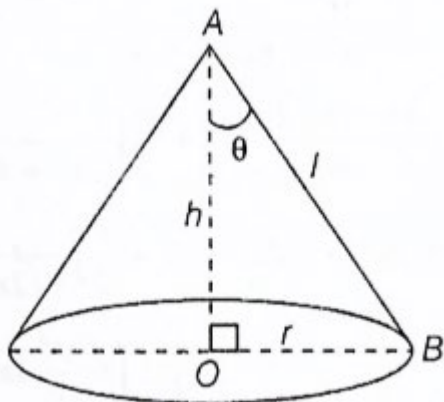
$$y_2 \cdot \sqrt{1-x^2} + y_1 \frac{1}{2\sqrt{1-x^2}}(-2x) = y_1$$

$$\frac{y_2(1-x^2) - xy_1}{\sqrt{1-x^2}} = y_1$$

$$y_2(1-x^2) - xy_1 = y_1 \sqrt{1-x^2}$$

$$y_2(1-x^2) - xy_1 = y$$

68. Consider the given figure.



In $\triangle AOB$

$$\tan 45^\circ = \frac{r}{h}$$

$$1 = \frac{r}{h}$$

$$r = h$$

Also,

$$l^2 = r^2 + h^2$$

$$l^2 = 2h^2$$

$$l = \sqrt{2}h$$

Therefore, the lateral surface area,

$$S = \pi r l$$

$$= \pi h \sqrt{2}h$$

$$= \sqrt{2}\pi (20.025)^2$$

$$= 401\sqrt{2}\pi$$

69. The rate of change of radius is,

$$\frac{dr}{dt} = 0.04 \text{ cm/sec}$$

The volume of the sphere is given by,

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

The surface area of the sphere is given as,

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

Therefore,

$$\frac{\frac{dV}{dt}}{\frac{dS}{dt}} = \frac{4\pi r^2 \frac{dr}{dt}}{8\pi r \frac{dr}{dt}}$$

$$\frac{dV}{dS} = \frac{r}{2}$$

$$= \frac{10}{2}$$

$$= 5 \text{ cm}$$

70. Consider the function

$$f(x) = a \cdot \log|x| + bx^2 + x$$

Differentiate with respect to x

$$f'(x) = a \frac{1}{|x|} \cdot \frac{x}{|x|} + 2bx + 1$$

$$= \frac{ax}{|x|^2} + 2bx + 1$$

It is given that $x = -1$ is one of the extremity of $f(x)$

$$f'(-1) = 0$$

$$\frac{a(-1)}{1} + 2b(-1) + 1 = 0$$

$$-a - 2b + 1 = 0$$

Also,

$$f'(-2) = 0$$

$$a \frac{(2)}{4} + 2b(2) + 1 = 0$$

$$a + 8b + 2 = 0$$

On solving both the equations,

$$a = 2$$

$$b = -\frac{1}{2}$$

Therefore,

$$(a, b) = \left(2, -\frac{1}{2}\right)$$

71. Consider the function.

$$f(x) = x^3 + px^2 + qx$$

And,

$$f(0) = f(2)$$

$$0 = 2^3 + 4p + 2q$$

$$2p + q + 4 = 0$$

Now,

$$f'(x) = 3x^2 + 2px + q$$

And,

$$f'\left(1 + \frac{1}{\sqrt{3}}\right) = 0$$

$$3\left(1 + \frac{1}{\sqrt{3}}\right)^2 + 2p\left(1 + \frac{1}{\sqrt{3}}\right) + q = 0$$

$$2p\left(1 + \frac{1}{\sqrt{3}}\right) + q + 4 + \frac{6}{\sqrt{3}} = 0$$

On solving both the equations,

$$p = -3$$

$$q = 2$$

Therefore,

$$\begin{aligned} p^2 + q^2 &= (-3)^2 + 2^2 \\ &= 9 + 4 \\ &= 13 \end{aligned}$$

72. Consider the integral

$$I = \int \frac{x^2}{2x^2 + 6\sqrt{2}x + 9} dx$$

It is solved as,

$$\begin{aligned} I &= \int \frac{\frac{1}{2}(2x^2 + 6\sqrt{2}x + 9) - \left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \int \frac{1}{2} dx - \int \frac{\left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - \int \frac{\left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \end{aligned}$$

Now,

$$3\sqrt{2}x + \frac{9}{2} = A(4x + 6\sqrt{2}) + B$$

Compare the coefficients of like terms.

$$4A = 3\sqrt{2}$$

$$A = \frac{3\sqrt{2}}{4}$$

And,

$$6\sqrt{2}A + B = \frac{9}{2}$$

$$B = -\frac{9}{2}$$

Therefore,

$$3\sqrt{2}x + \frac{9}{2} = 3\frac{\sqrt{2}}{4}(4x + 6\sqrt{2}) + \frac{9}{4}$$

Hence,

$$\begin{aligned} I &= \frac{x}{2} - \int \frac{3\frac{\sqrt{2}}{4}(4x + 6\sqrt{2}) + \frac{9}{4}}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - 3\frac{\sqrt{2}}{4} \int \frac{(4x + 6\sqrt{2})}{2x^2 + 6\sqrt{2}x + 9} dx + \frac{9}{4} \int \frac{1}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - \frac{3\sqrt{2}}{4} \log|2x^2 + 6\sqrt{2}x + 9| + \frac{9}{2} \int \frac{1}{(\sqrt{2}x + 3)^2} dx \\ &= \frac{x}{2} - \frac{3\sqrt{2}}{4} \log|(\sqrt{2}x + 3)^2| + \frac{9}{2} \frac{(\sqrt{2}x + 3)^{-2+1}}{-2+1} \cdot \frac{1}{\sqrt{2}} + c' \end{aligned}$$

Solve further,

$$\begin{aligned}
 I &= \frac{x}{2} - \frac{3\sqrt{2}}{2} \log \left| \left(\sqrt{2x+3} \right) \right| + \frac{9}{2\sqrt{2}} \frac{\left(\sqrt{2x+3} \right)^{-1}}{-1} + c' \\
 &= \frac{1}{2\sqrt{2}} \left[\left(\sqrt{2x+3} \right) - 6 \log \left| \sqrt{2x+3} \right| - \frac{9}{\sqrt{2x+3}} \right] + c
 \end{aligned}$$

73. Consider the given integral.

$$I = \int \frac{\sqrt{1-x^2} \sin^{-1} x + x}{\sqrt{1-x^2}} dx$$

It is solved as,

$$\begin{aligned}
 I &= \int \sin^{-1} x dx + \int \frac{x}{\sqrt{1-x^2}} dx \\
 &= \sin^{-1} x \cdot x - \int \frac{1}{\sqrt{1-x^2}} \cdot x dx + \int \frac{x}{\sqrt{1-x^2}} dx + c \\
 &= x \sin^{-1} x + c
 \end{aligned}$$

74. Consider the equation.

$$\int x^3 e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$$

Simplify the above equation as follows

$$x^3 \cdot \frac{e^{2x}}{2} - \int 3x^2 \cdot \frac{e^{2x}}{2} dx = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{2} \left[x^2 \frac{e^{2x}}{2} - \int 2x \cdot \frac{e^{2x}}{2} dx \right] = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \frac{3}{2} \int x \cdot e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \frac{3}{2} \left[x \frac{e^{2x}}{2} - \int 1 \cdot \frac{e^{2x}}{2} dx \right] = \frac{e^{2x}}{8} f(x) + c$$

Simplify further,

$$\frac{e^{2x}}{8} [4x^3 - 6x^2 + 6x - 3] + c_1 = \frac{e^{2x}}{8} f(x) + c$$

Therefore, on comparison

$$f(x) = 4x^3 - 6x^2 + 6x - 3$$

$$1 = 4x^3 - 6x^2 + 6x - 3$$

$$4x^3 - 6x^2 + 6x - 4 = 0$$

$$2(x^3 - 1) - 3x(x - 1) = 0$$

Solve further,

$$2(x - 1)(x^2 + x + 1) - 3x(x - 1) = 0$$

$$(x - 1)(2x^2 + 2x + 2 - 3x) = 0$$

$$(x - 1)(2x^2 - x + 2) = 0$$

$x = 1$ is a real root.

The sum of non-real complex roots from the above equation is $\frac{1}{2}$

75. Consider the integral.

$$\begin{aligned} I &= \int \sin^5 x \cos^5 x dx \\ &= \int \sin^4 x \sin x \cos^5 x dx \\ &= \int (1 - \cos^2 x)^2 \cos^5 x dx \end{aligned}$$

Let,

$$\cos x = t$$

$$-\sin x dx = dt$$

Therefore,

$$\begin{aligned} I &= \int (1 - t)^2 t^5 (-dt) \\ &= - \left[\frac{t^{10}}{10} - 2 \frac{t^8}{8} + \frac{t^6}{6} \right] + c \\ &= - \frac{t^6}{60} [6t^4 - 15t^2 + 10] + c \\ &= - \frac{\cos^6 x}{60} [6\cos^4 x - 15\cos^2 x + 10] + c \end{aligned}$$

Solve further,

$$\begin{aligned}
 I &= -\frac{\cos^6 x}{60} \left[6\sin^4 x - 12\sin^2 x + 6 - 15 + 15\sin^2 x + 10 \right] + c \\
 &= -\frac{\cos^6 x}{60} \left[6\sin^4 x - 3\sin^2 x + 1 \right] + c
 \end{aligned}$$

76. Consider the limit.

$$y = \lim_{x \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \frac{n^2}{(n+3)^3} + \dots + \frac{1}{125n} \right]$$

It is solved as,

$$\begin{aligned}
 y &= \lim_{x \rightarrow \infty} \left[\frac{n^2}{(n+0)^3} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{n^2}{(n+4n)^3} \right] \\
 &= \lim_{x \rightarrow \infty} \sum_{r=0}^{4n} \frac{n^2}{(n+r)^3}
 \end{aligned}$$

Let,

$$\frac{r}{n} = x$$

$$\frac{1}{n} = dx$$

Therefore,

$$\begin{aligned}y &= \int_0^4 \frac{dx}{(1+x)^3} \\&= \left[\frac{(1+x)^{-2}}{-2} \right]_0^4 \\&= -\frac{1}{2} \left[\frac{1}{5^2} - 1 \right] \\&= \frac{12}{25}\end{aligned}$$

77. Consider the integral.

$$I = \int_{e^{-1}}^{e^2} \left| \frac{\log x}{x} \right| dx$$

It can be simplified as,

$$I = \int_{e^{-1}}^1 \frac{\log x}{x} dx + \int_1^{e^2} \frac{\log x}{x} dx$$

Let,

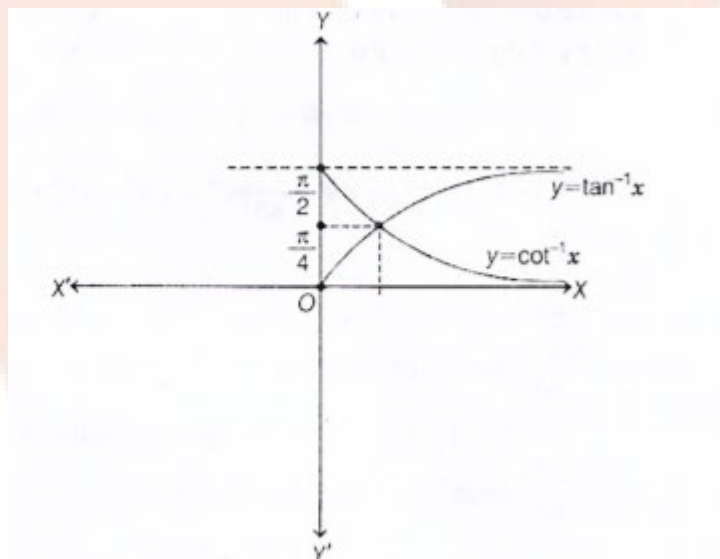
$$\log x = t$$

$$\frac{dx}{x} = dt$$

Thus,

$$\begin{aligned} I &= \int_{-1}^0 t dt + \int_0^1 t dt \\ &= \left[\frac{t^2}{2} \right]_{-1}^0 + \left[\frac{t^2}{2} \right]_0^1 \\ &= -\frac{1}{2}(0-1) + \frac{1}{2}(1-0) \\ &= \frac{1}{2} \end{aligned}$$

78. Consider the graph of $y = \tan^{-1}x$ and $y = \cot^{-1}x$.



The required area is calculated as,

$$\begin{aligned} A &= 2 \left[\frac{\pi}{4} - \int_0^1 \tan^{-1} x dx \right] \\ &= \frac{\pi}{2} - 2 \left[\left(x \tan^{-1} x \right)_0^1 - \int_0^1 \frac{x}{1+x^2} dx \right] \\ &= \frac{\pi}{2} - 2 \left[\frac{\pi}{4} - \frac{1}{2} \left[\log_e (1+x^2) \right]_0^1 \right] \\ &= \frac{\pi}{2} - \frac{\pi}{2} + \log_e 2 \end{aligned}$$

Solve further,

$$A = \log_e 2$$

79. The equation of family of parabolas with focus and the x axis is,
 $y^2 = 4a(x - a)$

Differentiate with respect to x .

$$2y \frac{dy}{dx} = 4a$$

$$a = \frac{y}{2} \frac{dy}{dx}$$

Therefore,

$$y^2 = 2y \frac{dy}{dx} \left(x - \frac{y}{2} \right) \frac{dy}{dx}$$

$$y^2 = y \frac{dy}{dx} (2x - y) \frac{dy}{dx}$$

$$y \left(\frac{dy}{dx} \right)^2 - 2x \left(\frac{dy}{dx} \right) + y = 0$$

Hence, the order of the differential equation is 1.

80. Consider the differential equation.

$$yy' = x \left[\frac{y^2}{x^2} + \frac{\phi \left(\frac{x^2}{y^2} \right)}{\phi' \left(\frac{x^2}{y^2} \right)} \right]$$

The above equation can be written as,

$$\frac{y}{x} \frac{dy}{dx} = \frac{y^2}{x^2} + \frac{\phi \left(\frac{x^2}{y^2} \right)}{\phi' \left(\frac{x^2}{y^2} \right)}$$

Let,

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Therefore,

$$v \left(v + x \frac{dv}{dx} \right) = v^2 + \frac{\phi(v^2)}{\phi'(v^2)}$$

$$vx \frac{dv}{dx} = \frac{\phi(v^2)}{\phi'(v^2)}$$

$$\int v \frac{\phi'(v^2)}{\phi(v^2)} dv = \int \frac{dx}{x}$$

$$\frac{1}{2} \log_e \phi(v^2) = \log_e x + \log_e (\sqrt{c})$$

Solve further,

$$\phi(v^2) = cx^2$$

$$\phi\left(\frac{y^2}{x^2}\right) = cx^2$$

